

Supplementary for An Efficient Insect-inspired Approach for Visual Point-goal Navigation

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I. MODEL DETAILS

More details of the user-defined parameters in our insect-inspired model are summarised in Table S1. We did not attempt to optimise the model, as α was set to be the highest possible learning rate without any tuning, and most other parameters were chosen to be the same as in our earlier work [1], where they were somewhat optimised but for the different model. Note τ_d and τ_c were set to different values in the two simulators because of the distinct low-level motion control mechanisms (§II-B3).

TABLE S1
MORE DETAILS OF OUR INSECT-INSPIRED MODEL.

variable	meaning	Habitat	iGibson	remarks
N_{PN}	number of PNs	1089	1089	determined by preprocessed image size [1]
$N_{PN/KC}$	number of PNs per KC	10	10	approximating real PN-KC connectivity [1]
N_{KC}	number of KCs	32000	32000	optimised in [2] for our earlier work [1]
k	number of activated KCs	320	320	k/N_{KC} approximating real KC sparsity [1]
α	learning rate of KC-MBON weights	1	1	highest possible rate
τ_d	epochs between visual and collision-related inputs	1	6	minimal hand-tuning
τ_c	epochs of escape	$2 \left\lfloor \frac{ \gamma_c - 90^\circ }{10^\circ} \right\rfloor$	15	minimal hand-tuning
F_c	collision threshold (N)	—	100	minimal hand-tuning [1]

More details of the seven recent point-goal navigation models cited in Table II are summarised in Table S2. Note HM3D is the dataset of scene reconstructions curated by [3], which includes Gibson and Matterport3D (MP3D) datasets as subsets. As [4] used a different simulator with a different dataset of scene reconstructions. The performance in Fig. 2 is inflated by a factor of 1.3 from the value in this table. The factor is calculated as the ratio of the different results of the same model in Gibson and MP3D scenes reported in Table 2 of [5].

TABLE S2
MORE DETAILS OF THE SEVEN MODELS IN TABLE II.

Model	Camera	Vision Model	RL /SLAM Model	Training Scenes	Training Epochs	SR	SPL	Reference
S1	RGB	CNN	ORB-SLAM2, D* Lite	SunCG, MP3D	5×10^6	0.090	0.047	Table 3 & 11 [4]
S2	RGBD	CNN	inherited from S1	Gibson 4+, MP3D	7.5×10^7	0.62	0.51	Table 2 [5]
V1	noisy RGB	ResNet18	inherited from E2	Gibson 4+	1.47×10^7	0.79	0.61	Table 2 [6]
V2	noisy RGBD	ResNet50	inherited from E2	Gibson 0+	7.5×10^9	0.960	0.766	Table 1 [7]
E1	RGB	same as S2	PPO	same as S2	7.5×10^7	0.64	0.46	Table 2 [5]
E2	RGB	SE-ResNeXt50	DDPPO	Gibson 2+, MP3D	2.5×10^9	0.991	0.929	Table 2 [8]
E3	RGB	ResNet50	inherited from E2	HM3D	1.5×10^9	1.00	0.90	Table 2 [3]

II. SUPPLEMENTARY FIGURES

See Fig. S1-S3.

REFERENCES

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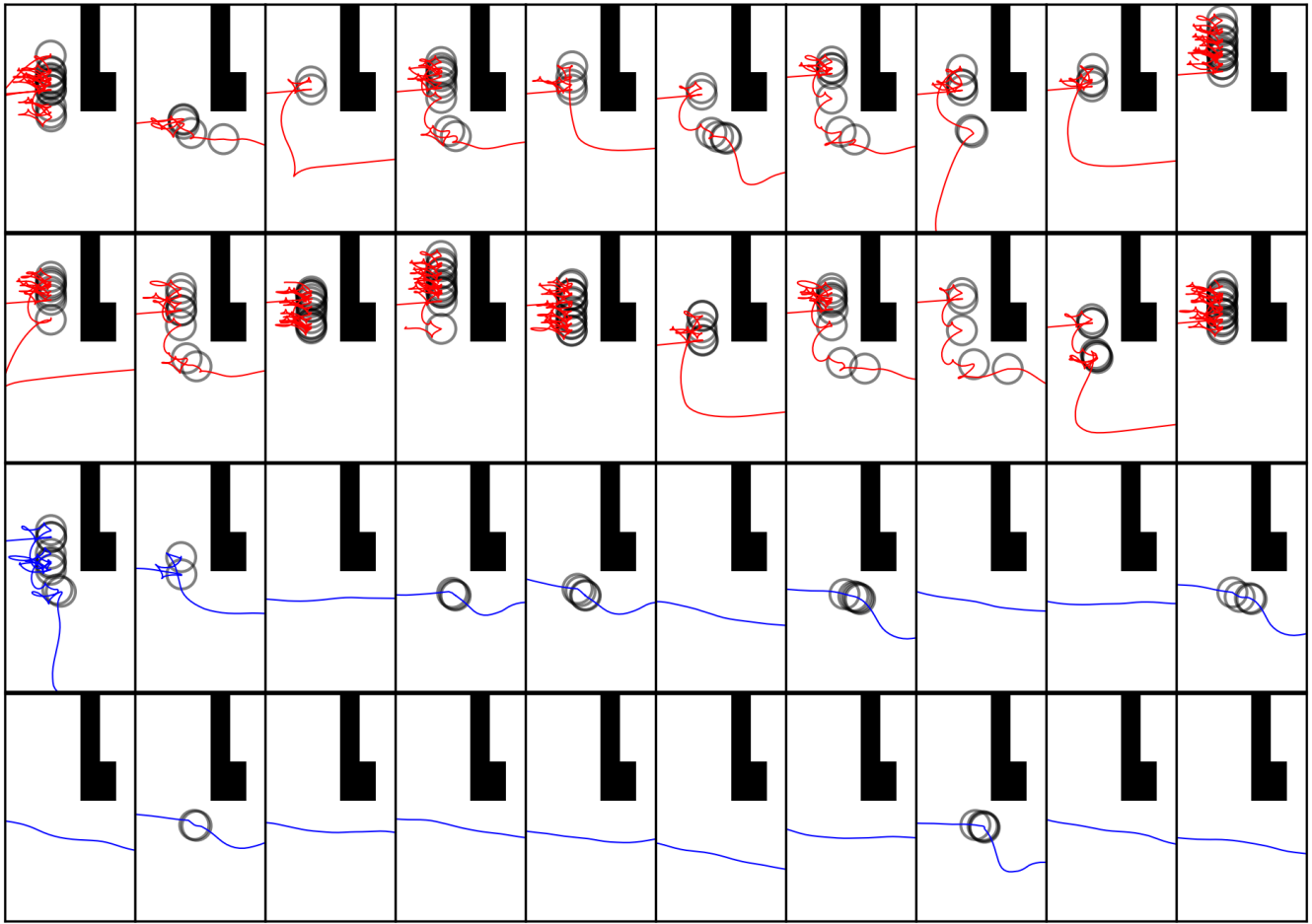


Fig. S1. Zoom-in plots of the area enclosed in the rectangle with dashed boundaries in Fig. 4. The black circles here denote where collisions were detected.

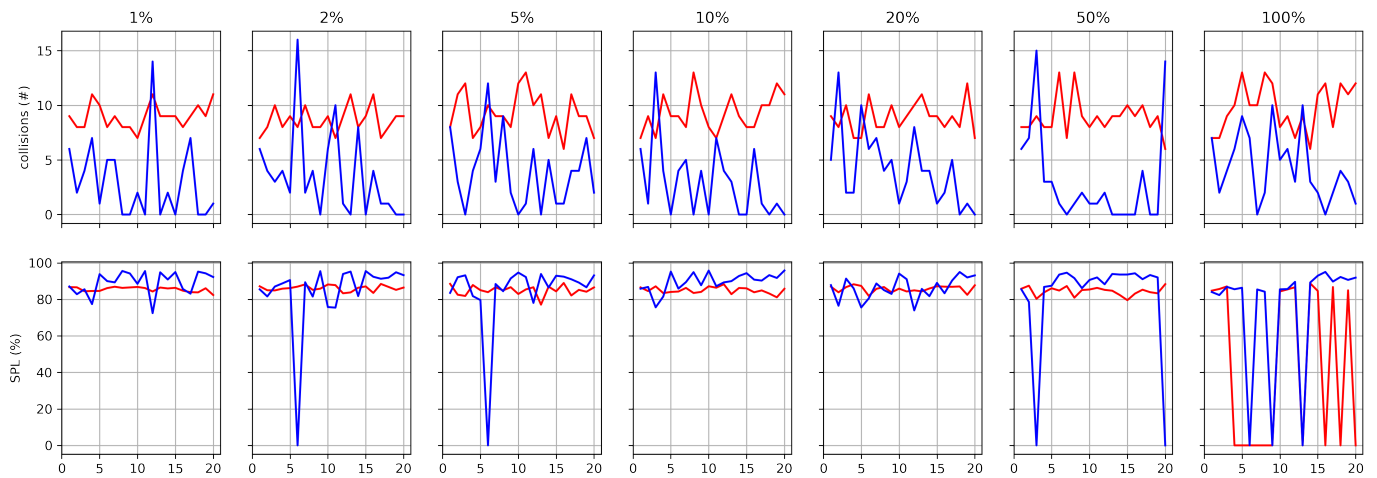


Fig. S2. The effect of motor noise on performance, across trials within an episode. Blue is the full model, red is ablated (odometry-collision).

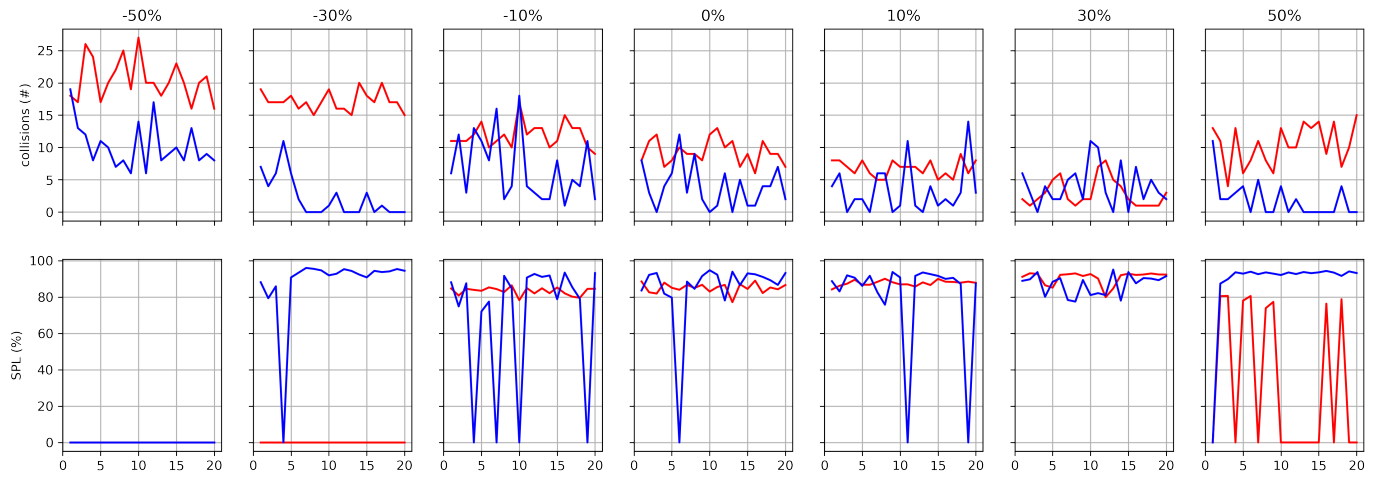


Fig. S3. The effect of motor bias on performance, across trials within an episode. Blue is the full model, red is ablated (odometry-collision).